

2 Assignment 3



Question 1: Decision Trees

The training data for a certain binary classification task is visualized in Figure ???. The feature space is $[0, 1]^2$.

You may treat the crosses ($\times$) as class +1 and the dots ($\bullet$) as class -1 when convenient.

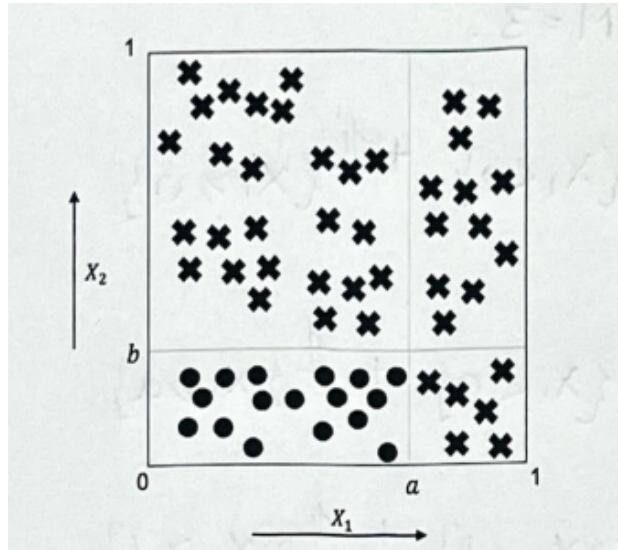


Figure 1: Data

Task: Come up with a decision tree that would fit the training data *perfectly*. Clearly specify:

- The feature used at each split
- The threshold value
- The class assigned to each leaf node

very split focuses on a small random feature

Question 2: Random Forests

In your own words, explain what a **Random Forest** is.

Your answer should address the following:

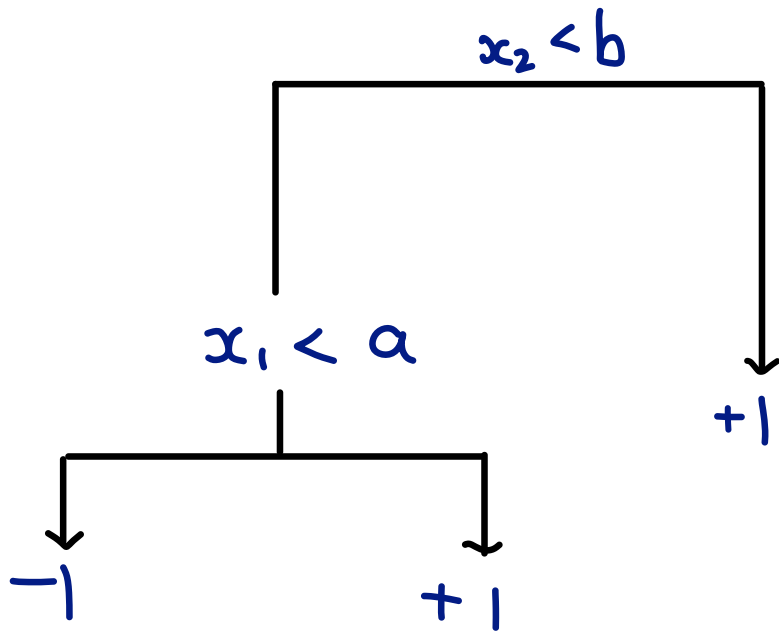
- Why is it called a *forest*?
- Why is it called *random*?
- How does randomness help improve performance compared to a single decision tree?

Question 3: Ensemble Methods

- What do you mean by an **ensemble method** in machine learning?
- Explain the motivation behind using ensemble methods.
- Would a model that combines the predictions of multiple decision trees be called an ensemble method? Justify your answer.

Q.1

TREE :



Feature :

First split : x_2 Second split : x_1

Thresholds:

used a & b respectively for the split because a boundary is visible that separates the 2 classes at both points.

Class:

$+1$ for $\times$ & -1 for $\bullet$ at each leaf node

Q.2

(an algorithm therein)

- Random forest refers to an ensemble method in ML which is kind of an extrapolation of the Decision Tree method, where instead of relying on a single tree, which can be unstable, we use multiple decision trees parallelly. (also why it's called a 'forest') It is notably 'random' because each of these trees & their splits in turn, focus on small random features to make their classifications.
- This is better than using a single tree because it allows the algorithm to tap into multiple possibilities of splits with small variations across the data without putting heavy load on a single complex tree which would be unstable and would risk overfitting.

A good metaphor I came across was considering a random forest to be like a panel of experts; who are much more valuable to consult than just a single person of a similar expertise level.

Q.3

- 'Ensemble Methods' are ML techniques in which we use multiple models simultaneously to train on data, such that it can use the collective strength of all the models to give a more reliable and accurate.
- The reason we turn to this method is due to having multiple models that approach the data varyingly, the errors average out and reduce, which enables the machine to better generalise unseen data, fulfilling our purpose more reliably and accurately.
- Yes, it is considered an ensemble method as essentially a multi-layered process forms basis of the model. The effectiveness of this although is highly subjective as it depends on the parameters of each of the trees (& how much they vary). As largely same or largely different metrics defeat the purpose of reducing avg. error. Slightly varied parameters (as in Random Forests) are ideal.

Question 4: Confusion Matrix

Consider a binary classification model used to detect a disease. The model is evaluated on a test set of 1000 patients and produces the following confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	180	20
Actual Negative	70	730

1. Identify the values of:

- True Positives (TP) $\rightarrow 180$
- False Positives (FP) $\rightarrow 70$
- True Negatives (TN) $\rightarrow 730$
- False Negatives (FN) $\rightarrow 20$

2. Compute the following performance metrics:

- Accuracy $\Rightarrow 91\%$
- Precision $\Rightarrow 72\%$
- Recall (Sensitivity) $\Rightarrow 90\%$
- Specificity $\Rightarrow 91.25\%$
- F1-score $\Rightarrow \frac{2 \times 0.72 \times 0.90}{0.162} = 80\%$

$$\text{Accuracy} \rightarrow \frac{\text{True}}{\text{Total}} = \frac{180 + 730}{180 + 730 + 70 + 20} = \frac{910}{1000} = 0.91$$

$$\text{Precision} : \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{180}{250} = \frac{18}{25}$$

$$\text{Recall} : \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{180}{200} = 0.90$$

$$\text{Specificity} : \frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{730}{800}$$

3. Suppose the cost of a **false negative** is much higher than a false positive. Which metric would you prioritize while evaluating the model? Explain why.
4. If the classification threshold is lowered, which entries of the confusion matrix are most likely to increase? Justify your answer qualitatively.
5. Can two classifiers have the same accuracy but very different confusion matrices? Explain with reasoning.

3.) In such a scenario we need to pay attention to 'Recall' as it essentially measures the % of the positives that are detected with $\frac{\text{TP}}{\text{TP} + \text{FN}}$, $\therefore$ higher the Recall, lower the no. of FN which is our goal as missing a detection has a high cost. F2 scores could also help weigh Precision & Recall.

4.) Likely that the false positives increase along with True Positives i.e. FPs & TPs due to threshold being lowered borderline cases will also fall through the classifier and that is more likely to cause Type 1 errors. If the threshold is lowered to a large extent then there will be a large decrease in both FN & TNs.

5.) Yes. Accuracy only measures how 'correct' the predictions are. Mathematically : $\frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$ denotes accuracy. $\therefore$ While the numbers could match the FP and FN could be vastly different, indicating differently inclined models & higher accuracies can be misleading. eg. A case with 90% No and 10% Yes, if a model suggests 'No' 100% times, its accuracy is 90%. but it has 0 Recall, which is bad.